HDOC - 71st day

DYNAMIC MATRIX OPERATIONS USING POINTERS

Name – Piyush Negi Sap_Id – 590052185

Problem Statement

Write a C program to dynamically allocate two matrices and perform addition, subtraction, and multiplication using pointer notation. Implement each operation as a separate function and release all allocated memory properly.

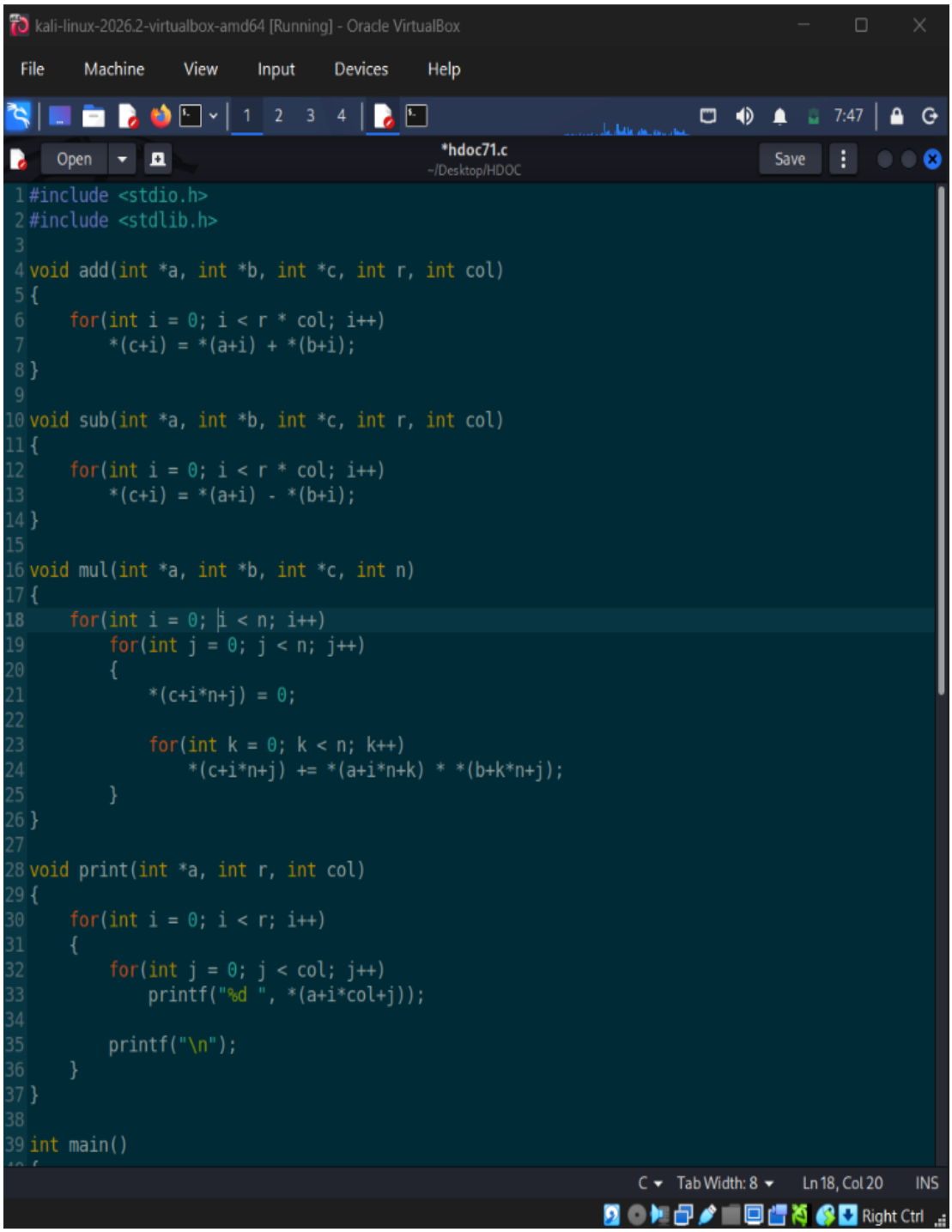
Algorithm

1. Start.
2. Read the size n of the square matrices.
3. Dynamically allocate memory for matrices a, b and c using malloc().
4. Read the elements of the first and second matrices using pointer notation.
5. Call add() to add corresponding elements of the two matrices.
6. Call sub() to subtract corresponding elements of the two matrices.
7. Call mul() to multiply the two matrices using pointer notation.
8. Display the result after each matrix operation.
9. Release all dynamically allocated memory using free().
10. Stop.

Test Case Table

S. No.	Input	Expected Output	Actual Output	Result
1	n = 2 A = 1 2 / 3 4 B = 5 6 / 7 8	Add: 6 8 / 10 12 Sub: -4 -4 / -4 -4 Mul: 19 22 / 43 50	Add: 6 8 / 10 12 Sub: -4 -4 / -4 -4 Mul: 19 22 / 43 50	Pass
2	n = 3 A = 1 2 3 / 4 5 6 / 7 8 9 B = 10 11 12 / 13 14 15 / 16 17 18	Add: 11 13 15 / 17 19 21 / 23 25 27 Sub: all -9 Mul: 84 90 96 / 201 216 231 / 318 342 366	Add: 11 13 15 / 17 19 21 / 23 25 27 Sub: all -9 Mul: 84 90 96 / 201 216 231 / 318 342 366	Pass

C Program




```c
#include <stdio.h>
#include <stdlib.h>

void add(int *a, int *b, int *c, int r, int col)
{
    for(int i = 0; i < r * col; i++)
        *(c+i) = *(a+i) + *(b+i);
}

void sub(int *a, int *b, int *c, int r, int col)
{
    for(int i = 0; i < r * col; i++)
        *(c+i) = *(a+i) - *(b+i);
}

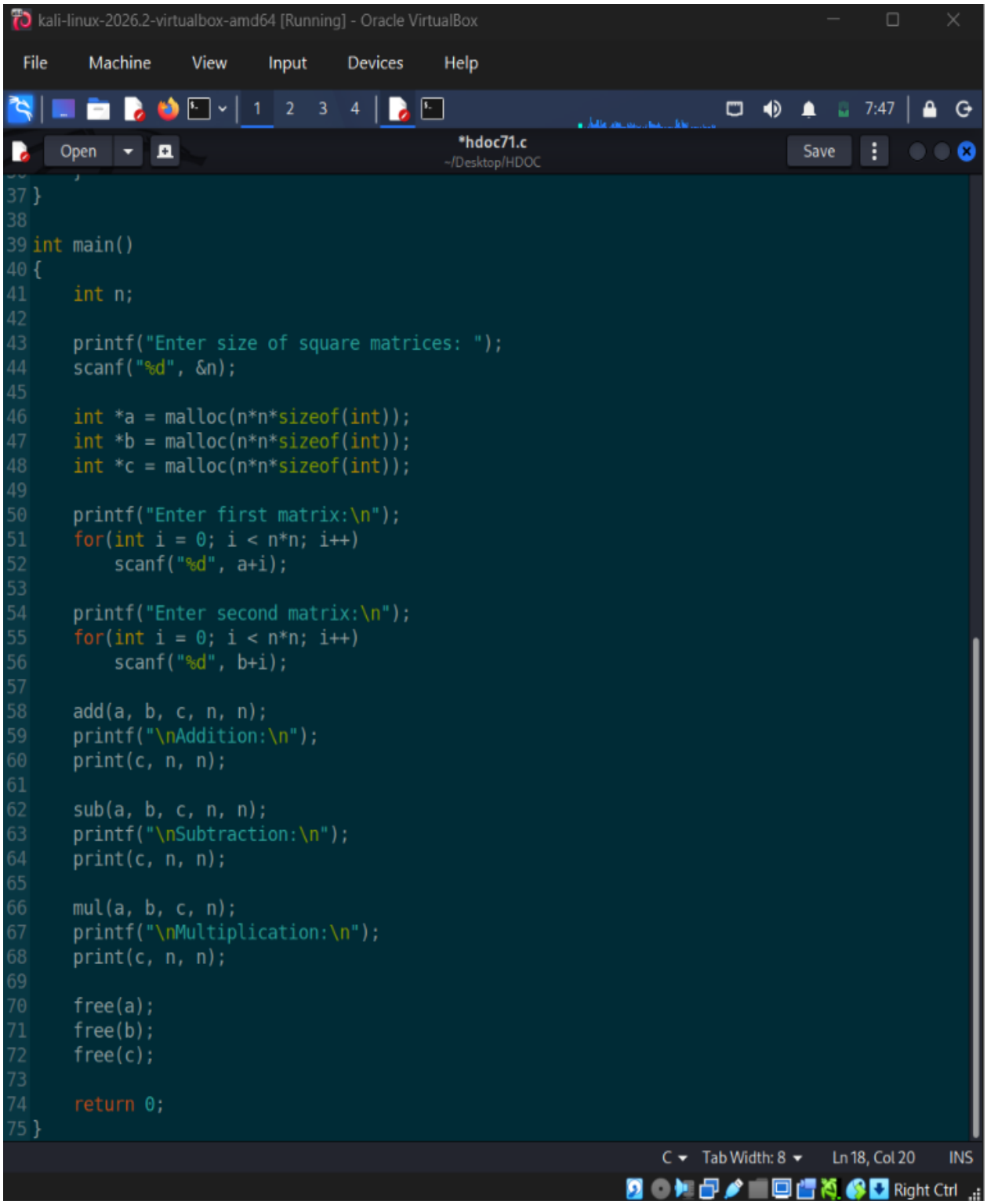
void mul(int *a, int *b, int *c, int n)
{
    for(int i = 0; i < n; i++)
        for(int j = 0; j < n; j++)
        {
            *(c+i*n+j) = 0;

            for(int k = 0; k < n; k++)
                *(c+i*n+j) += *(a+i*n+k) * *(b+k*n+j);
        }
}

void print(int *a, int r, int col)
{
    for(int i = 0; i < r; i++)
    {
        for(int j = 0; j < col; j++)
            printf("%d ", *(a+i*col+j));

        printf("\n");
    }
}

int main()
```


kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
7:47
Open
*hdoc71.c
~/Desktop/HDOC
Save
}
int main()
{
int n;
printf("Enter size of square matrices: ");
scanf("%d", &n);
int *a = malloc(n*n*sizeof(int));
int *b = malloc(n*n*sizeof(int));
int *c = malloc(n*n*sizeof(int));
printf("Enter first matrix:\n");
for(int i = 0; i < n*n; i++)
scanf("%d", a+i);
printf("Enter second matrix:\n");
for(int i = 0; i < n*n; i++)
scanf("%d", b+i);
add(a, b, c, n, n);
printf("\nAddition:\n");
print(c, n, n);
sub(a, b, c, n, n);
printf("\nSubtraction:\n");
print(c, n, n);
mul(a, b, c, n);
printf("\nMultiplication:\n");
print(c, n, n);
free(a);
free(b);
free(c);
return 0;
}
C Tab Width: 8 Ln 18, Col 20 INS
Right Ctrl

Compilation and Execution

Compilation command: gcc hdoc71.c -o hdoc71

Execution command: ./hdoc71

Execution Screenshot:

